

Cherenkov diffraction radiation

From bunch-length diagnostics to optical microbunching

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Setup

$$K = 60 \text{ MeV}, \quad Q = 1 \text{ nC}, \quad \sigma_z = 1 \text{ mm}, \quad \sigma_t = 3.33576 \text{ ps}.$$

σ_z : rms bunch length; $\sigma_t = \sigma_z/v$: rms duration.

Case 1: full bunch length

Use the coherent spectral roll-off to infer the rms length of the bunch envelope.

The GHz band is the simpler starting point: longer wavelengths relax mesh resolution and beam–radiator gap requirements.

Case 1 establishes the experimental and numerical workflow. A bunch length measurement could be a simple case for the 2027 AWA measurement campaign. Too simple? Case 2 extends it to finer structure and our goal.

Case 2: fine structure / microbunching

The main application: detect small-scale longitudinal structure within the same millimetre-long bunch.

The $\lambda_0 = 0.5 \text{ }\mu\text{m}$ band tests optical-scale bunching against the shot-noise background.

What motivates the $0.5 \mu\text{m}$ scale? (J Power question)

Illustrative density estimate

$$Q = 1 \text{ nC},$$

$$V = (1 \text{ mm})^3,$$

$$N = Q/e = 6.2415 \times 10^9,$$

$$n = 6.2415 \times 10^{18} \text{ m}^{-3},$$

$$d_{\text{cube}} = n^{-1/3} = 0.5431 \mu\text{m}.$$

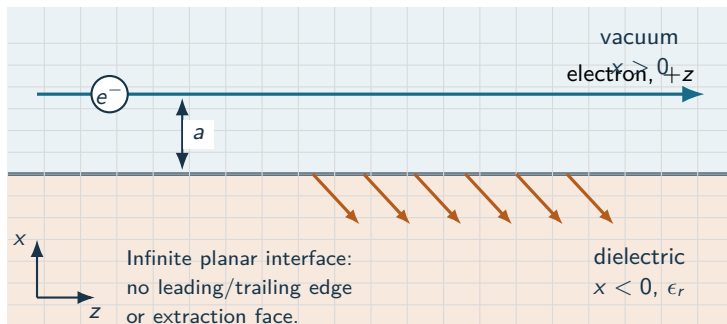
The uniform box defines this estimate; it is not the volume of the Gaussian bunch with $\sigma_z = 1 \text{ mm}$. d_{cube} is a volume-per-electron scale.

Note

The density scale is numerically correct, but it does **not** select a radiation line at $\lambda_0 = d_{\text{cube}}$. The mean-spacing estimate motivates an optical scale to investigate. Studying longitudinal modulation near $0.5 \mu\text{m}$ requires sensitivity around 600 THz.

- Independent random electrons give broadband microscopic noise.
- Microbunching adds correlated longitudinal structure, measured through the bunching factor.
- The single-electron response and optical collection determine how strongly this structure is observed.

Planar reference geometry



a : electron–surface gap; x : surface normal; z : beam axis; y : direction out of the drawing. Orange arrows show energy entering the dielectric.

A simple “1-D” calculation

The electron follows $(x, y, z) = (a, 0, vt)$. Each Fourier component has phase $e^{ik_y y + ik_z z - i\omega t}$:

$$\omega = 2\pi f, \quad k_z = \frac{\omega}{v}.$$

Time and y are Fourier transformed; translation along the trajectory fixes k_z . For each (ω, k_y) , solve in x :

$$\kappa = \sqrt{k_y^2 + \frac{\omega^2}{\gamma^2 v^2}}, \quad k_{x,d}^2 = \epsilon_r \frac{\omega^2}{c^2} - k_z^2 - k_y^2.$$

In vacuum the interface field decays as $e^{-\kappa a}$. Real $k_{x,d}$ allows propagation into the dielectric. k_y : transverse Fourier wave number. At each radiation frequency, the “1-D” code numerically integrates the contributions from the relevant k_y components. κ : vacuum decay rate; $k_{x,d}$: normal propagation constant.

Planar spectral construction: [1], Sec. III, Eqs. (5)–(9); SI convention on the Fourier-normalization slide.

Single electron: energy flux and energy loss

Define $\mathcal{W}_1(f) = d^2 W_1 / (dz df)$: energy per path length and frequency, in $\text{J m}^{-1} \text{Hz}^{-1}$.

$$\mathcal{W}_1(f) = -\frac{1}{\pi} \int_{-\infty}^{\infty} dk_y \operatorname{Re} \left[(\hat{\mathbf{E}}_d \times \hat{\mathbf{H}}_d^*)_x \right] \quad (1)$$

The cross product is spectral Poynting flux. The minus sign selects energy entering $x < 0$. Subscript d : dielectric; hats: complex Fourier amplitudes; *: complex conjugation;

$$\mathcal{W}_1(f) = -\frac{q}{\pi} \int_{-\infty}^{\infty} dk_y \operatorname{Re} \hat{E}_{z,\text{ref}}(a, k_y, 2\pi f) \quad (2)$$

$q = -e$: electron charge. The reflected longitudinal field acts back on the electron; Eq. (2) measures its energy loss.

Interpretation: energy entering the lossless dielectric equals energy lost by the electron.

Fields: [1], Sec. III. Eqs. (1)–(2) are our SI Poynting/Parseval derivation; the $1/\pi$ normalization is explained in the appendix, not copied from [1].

From the bunch profile to the expected spectrum

The normalized profile $g(t)$ describes electron arrival times, with $\int g(t)dt = 1$:

$$F(f) = \int_{-\infty}^{\infty} g(t)e^{i2\pi ft} dt = \langle e^{i2\pi ft_j} \rangle.$$

Meaning: average the unit waves $e^{i2\pi ft}$ across the profile $g(t)$. F is the surviving complex amplitude per electron. $|F| \simeq 1$: waves align; $|F| \simeq 0$: they cancel.

For fixed N , independent arrival times from $g(t)$, and the same transverse track:

$$\langle \mathcal{W}_N \rangle = \underbrace{N\mathcal{W}_1}_{\text{individual intensities}} + \underbrace{N(N-1)|F|^2\mathcal{W}_1}_{\text{interference between electrons}} \quad (3)$$

What is averaged? Possible electron arrival times for the *same* N and $g(t)$. No averaging of measured bunches is required.

Eq. (3) supplies the expected signal and shot-noise reference from the profile alone. One actual bunch can fluctuate around this prediction.

[2], Eqs. (29)–(30). The next slide gives the exact spectrum for one specified set of arrival times.

One actual bunch versus the statistical prediction

If all arrival times t_j are known, sum their fields directly:

$$T_N(f) = \sum_{j=1}^N e^{i2\pi f t_j}, \quad \boxed{\mathcal{W}_N(f) = |T_N(f)|^2 \mathcal{W}_1(f)}.$$

T_N is the **dimensionless complex phase sum** for one actual bunch. Each term has magnitude one and phase $2\pi f t_j$; $|T_N|^2$ multiplies the single-electron spectrum, without averaging. Eq. (3) is useful when only the smooth profile is specified.

$|T_N|^2 = N + \sum_{j \neq l} e^{i2\pi f(t_j - t_l)}$. For independent arrivals, each of the $N(N-1)$ ordered pairs has expected phase product $\langle e^{i2\pi f(t_j - t_l)} \rangle = FF^* = |F|^2$.

$$\boxed{\mathcal{W}_{\text{fluct}} = (\langle |T_N|^2 \rangle - |\langle T_N \rangle|^2) \mathcal{W}_1 = N(1 - |F|^2) \mathcal{W}_1} \quad (4)$$

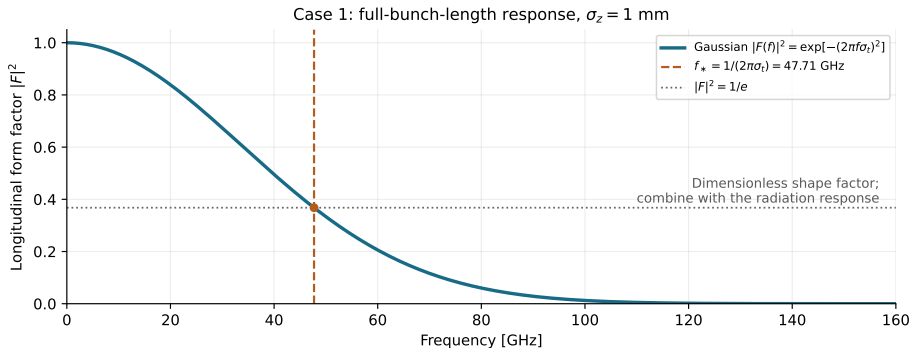
Here $\langle T_N \rangle = NF$. Eq. (4) describes field fluctuations about this mean.

For $F \simeq 0$, the smooth mean field cancels, but the expected intensity remains $N\mathcal{W}_1$: the shot noise.

For the Gaussian envelope used here, $|F|^2 = e^{-(2\pi f \sigma_t)^2}$.

[2], Eqs. (29)–(30) and Summary, pp. 125–126. The present Python spectra evaluate Eq. (3); a specified particle realization uses the direct phase sum.

Case 1: measuring the full bunch length

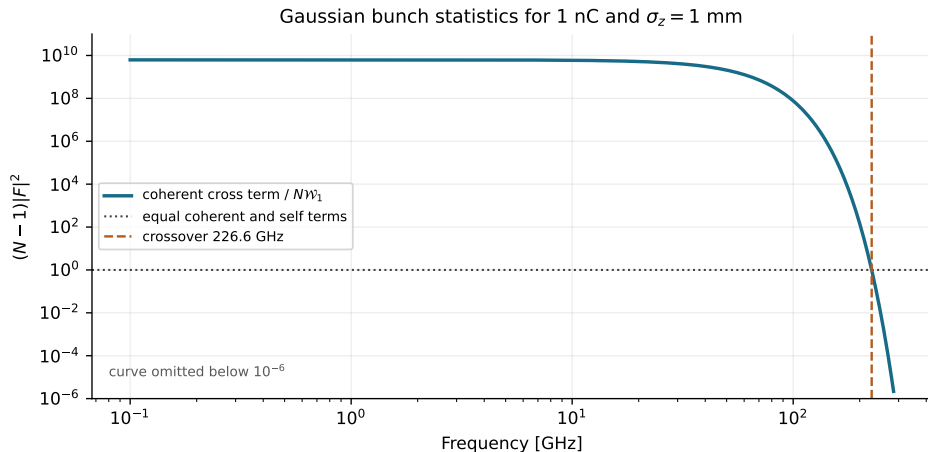


$$|F(f_*)|^2 = e^{-1}, \quad f_* = \frac{1}{2\pi\sigma_t} = 47.71 \text{ GHz}, \quad \sigma_z = \frac{v}{2\pi f_*}.$$

Fit the coherent spectral roll-off after accounting for the single-electron response and detector transfer function. Longer wavelengths make this a practical first validation case.

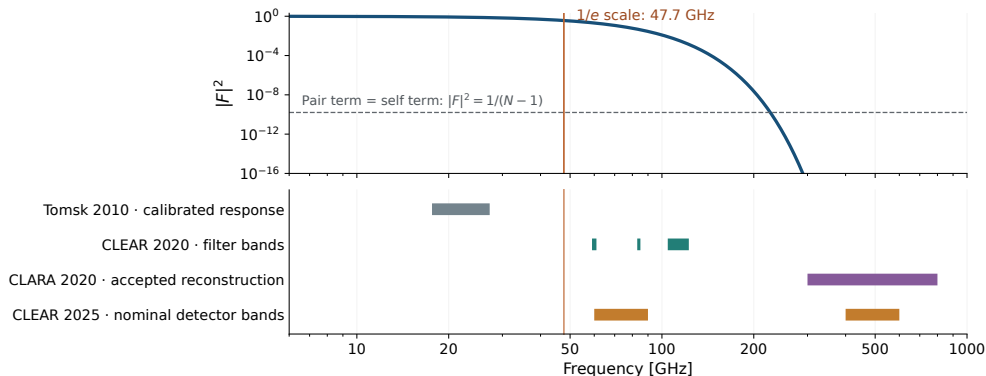
Form-factor diagnostics: [3], Sec. 3, Eqs. (21), (24)–(27), (33). The shared-arrival-time argument applies here to ChDR.

Case 1: coherent signal versus shot noise



The ratio is $(N - 1)|F|^2$, from Eq. (3). Interference and individual intensities become equal near 226.59 GHz. The bunch-length roll-off $f_* = 47.71$ GHz occurs earlier.

Published measurement bands overlap our GHz case



For our 1 mm rms bunch, **20–120 GHz** spans the spectral slope. CLEAR's 400–600 GHz channel targets shorter bunches.

Sources: [5]–[7]. Bars indicate detector/filter bands or the accepted reconstruction interval. The curve contains only the longitudinal form factor. Radiator and detector responses remain to be included.

Case 2: optical-scale structure within the bunch

For one bunch, define the **bunching factor**

$$b(f) = \frac{1}{N} \sum_{j=1}^N e^{i2\pi f t_j}, \quad \boxed{\mathcal{W}_N(f) = N^2 |b(f)|^2 \mathcal{W}_1(f)}.$$

$|b|$ measures phase alignment in that bunch; $F = \langle b \rangle$ describes the mean longitudinal profile.

What does $0.5 \mu\text{m}$ probe?

$\lambda_0 = c/f$ is the vacuum wavelength. A longitudinal modulation at period λ_b gives

$$\lambda_b = v/f = \beta\lambda_0.$$

At 60 MeV, this is approximately $0.5 \mu\text{m}$; the corresponding time period is 1.6678 fs.

What would indicate microbunching?

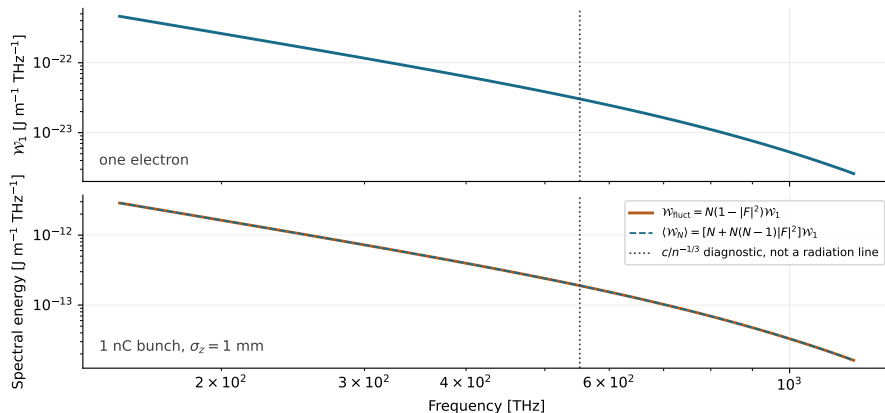
The smooth 1 mm Gaussian has negligible optical F . Its random bunching level is

$$\sqrt{\langle |b|^2 \rangle} \simeq N^{-1/2} = 1.266 \times 10^{-5}.$$

An excess or feature above this reference can indicate microstructure, after calibrating the radiator and detector response.

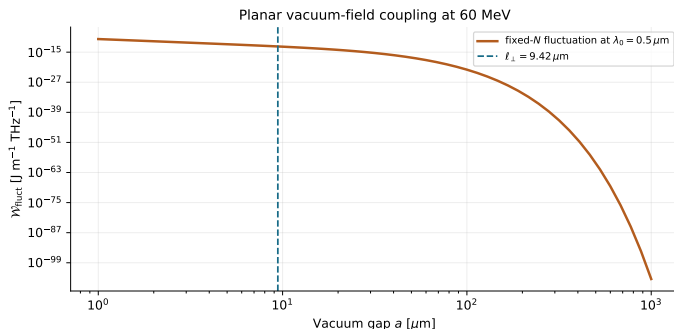
Phase summation: [3], Sec. 3.1; random-bunching baseline: [2]. This intrinsic statistical level is not a detector sensitivity limit.

Case 2: optical shot-noise reference spectrum



Smooth Gaussian reference: $K = 60$ MeV, $Q = 1$ nC, $\sigma_z = 1$ mm, $a = 10$ μm , real $\epsilon_r = 2.13$. At $\lambda_0 = 0.5$ μm , $\mathcal{W}_{\text{fluct}} = 1.542 \times 10^{-25}$ J m⁻¹ Hz⁻¹. Compare a future microbunched spectrum with this unmodulated baseline.

Case 2: the gap controls optical coupling



At $\lambda_0 = 0.5 \mu\text{m}$,

$$\ell_{\perp} = \frac{\gamma\beta\lambda_0}{2\pi} = 9.423 \mu\text{m}.$$

At the earlier illustrative $a = 1 \text{ mm}$, the least-evanescent exponential coupling factor is

$$\exp(-2a/\ell_{\perp}) = 6.644 \times 10^{-93}.$$

$\ell_{\perp}$ is the longest vacuum-field decay length at this frequency.

Optical coupling favors electrons close to the surface; use the actual transverse beam distribution to predict the participating charge.

Optical measurements: coupling at 60 MeV

Experiment	Electrons	Optical selection
CESR [8]	5.3 GeV	600 ± 10 nm as reported in figure captions
ATF2 [9]	1.3 GeV	700 nm centre, nominal 40 nm FWHM filter
CLEAR [10]	200 MeV	600 ± 10 nm, preliminary in-air result

These experiments establish incoherent optical ChDR for beam imaging or position. The reviewed sources contain no demonstration of optical ChDR microbunching diagnostics.

$$\ell_{\perp} = \frac{\gamma\beta\lambda_0}{2\pi} = 9.423 \mu\text{m} \quad (60 \text{ MeV}, \lambda_0 = 0.5 \mu\text{m}).$$

Our design constraint: a $10 \mu\text{m}$ centre gap would intercept a substantial part of a 1 mm rms beam. Optical coupling needs a consistent normal beam size and clearance.

[8] Secs. III–IV; [9] p. 2656 and filter specification; [10] pp. 662–663. CESR prose calls the filter “10 nm bandpass”, so its FWHM remains ambiguous. CLEAR’s large-gap in-air signal needs separate assessment against our vacuum model.

What this model can validate in IPPL

Direct comparisons

- complex fields at specified laboratory probes;
- TE/TM interface behavior and dielectric propagation angle;
- gap dependence and flux/work energy consistency;
- the single-electron spectral response before bunch statistics.

Route from reference to experiment

- include the finite prism and its optical extraction face;
- use measured, frequency-dependent complex $\epsilon(\omega)$;
- include transverse beam size and the distribution of gaps;
- calibrate detector bandwidth and collection optics.

Recommended sequence

Case 1: validate the coherent bunch-length signal. Case 2: add prescribed microbunching and compare its spectrum with the noise reference.

Moving sources in FDTD: [4], Sec. 4.7. Numerical checks use common field phases, spectral normalization and geometry.

Appendix: Fourier convention and the $1/\pi$ factor

For the electric field $\mathbf{U} = \mathbf{E}$ or magnetic field strength $\mathbf{U} = \mathbf{H}$, we use

$$\mathbf{U}(x, y, z, t) = \frac{1}{(2\pi)^2} \int d\omega dk_y \hat{\mathbf{U}}(x, k_y, \omega) e^{ik_y y + i\omega z/v - i\omega t}.$$

The transformed charge and current densities are $\hat{\rho} = (q/v)\delta(x - a)$ and $\hat{J}_z = q\delta(x - a)$. The Dirac delta δ locates the electron at $x = a$.

$$\frac{W_1}{L} = - \int dt dy (\mathbf{E}_d \times \mathbf{H}_d)_x = - \frac{1}{(2\pi)^2} \int d\omega dk_y \operatorname{Re}(\hat{\mathbf{E}}_d \times \hat{\mathbf{H}}_d^*)_x.$$

Parseval's identity converts time/position integration to a product of Fourier amplitudes. Real fields give equal contributions at positive and negative frequency. With $d\omega = 2\pi df$,

$$\underbrace{\frac{2(2\pi)}{(2\pi)^2}}_{f>0} = \frac{1}{\pi}.$$

Here L is trajectory length; $\hat{\mathbf{E}}$ has units V s, $\hat{\mathbf{H}}$ has units A s.

Local SI derivation of Eq. (1). Applying the same inverse transform to the induced work $-qE_{z,\text{ref}}$ gives Eq. (2). Field construction: [1].

Reproducibility and references

In this directory

- `make_figures.py`: regenerates the figures from `../chdr_1d.py`;
- `chdr_1d_presentation.tex`: TeXShop-ready Beamer source;
- `Makefile`: make figures then make pdf.

References

1. A. V. Tyukhtin, S. N. Galyamin and V. V. Vorobev, *Radiation of Charge Moving along Face of Inverted Prism*, [arXiv:2105.01111v2](#) (2022). Sec. III, Eqs. (5)–(9): planar field construction (Gaussian units).
2. R. A. Bosch and R. J. Bosch, *Derivation of Bunching for Poisson Statistics*, [FEL2009](#), [MOPC44](#), pp. 123–126. Eqs. (29)–(30) and p. 126: moments at *fixed* electron count.
3. B. Schmidt et al., *Longitudinal Bunch Diagnostics using Coherent Transition Radiation Spectroscopy*, [DESY 18-027](#) (2018). Sec. 3, Eqs. (21), (24)–(27), (33): phase summation and form factors.
4. A. Oskooi and S. G. Johnson, *Electromagnetic Wave Source Conditions*, in *Advances in FDTD Computational Electrodynamics*, eds. A. Taflove, A. Oskooi and S. G. Johnson (2013). [Chapter 4](#), Sec. 4.7: moving charges.

Measurement literature

5. A. P. Potylitsyn et al., *Coherent Cherenkov Radiation from a Short Bunch Passing near a Target and Possibility of a Bunch Length Diagnostics*, [IPAC2010, MOPE046](#). Experimental section: calibrated detector band and $\sigma_z = 1.2$ mm.
6. A. Curcio et al., *Noninvasive bunch length measurements exploiting Cherenkov diffraction radiation*, [PRAB 23, 022802 \(2020\)](#). Secs. IV–V: CLEAR channels and CLARA spectral reconstruction.
7. C. Davut et al., *Design and experimental verification of a bunch length monitor based on coherent Cherenkov diffraction radiation*, [PRResearch 7, 013193 \(2025\)](#). Tables II–III and Sec. V: detector bands and measured beam parameters.
8. R. Kieffer et al., *Generation of incoherent Cherenkov diffraction radiation in synchrotrons*, [PRAB 23, 042803 \(2020\)](#).
9. M. Bergamaschi et al., *Recent Results using Incoherent Cherenkov Diffraction Radiation for Non-Invasive Beam Diagnostics*, [IPAC2019, WEPGW077](#). Thorlabs FB700-40 specification, p. 395.
10. D. Alves et al., *Cherenkov Diffraction Radiation as a Tool for Beam Diagnostics*, [IBIC2019, THAO01](#).

Band comparison generator: `../.. /ideas-lit-reseach/compare_measurement_bands.py`. make figures regenerates the comparison and copies its slide version here.